October 14th:

-What do I want:

I want something I can run in my local hardware that when someone speaks, it transcribes the audio in real-time (or close to it). Then we can analyze the words being transcribed for keywords.

What kind of keywords?
(Command) followed by (Modifier) followed by (Identifier)

Command word options?

-Deliver

-Request

-Fetch

Modifier options?

-green

-red

-blue

-yellow

-grey

-black

-small

-medium

-large

Identifier options?
-screwdriver

-tool

-object

-wrench

Options:
-faster whisper:

https://github.com/SYSTRAN/faster-whisper

https://pypi.org/project/faster-whisper/

>model size options: ‘large-v3’ ‘base’ ‘small’ ‘medium’ ‘tiny’, (etc)

Oct 14 - First Successful Run:

#parameters: base, cpu, int8

(misc) PS C:\Users\mrrya\Documents\MREN403\misc> python -u "c:\Users\mrrya\Documents\MREN403\misc\voice.py"

Detected language 'en' with probability 0.983152

[0.00s -> 15.64s] Deliver, Request, Fetch, Green, Red, Blue, Yellow, Gray, Black, Small, Medium, Large,

[15.64s -> 19.64s] Screwdriver, Tool, Object, Ranch.

(misc) PS C:\Users\mrrya\Documents\MREN403\misc>

Oct 14 – Run 2
#large-v3 taking a long time to download, obviously

#used small model

Detected language 'en' with probability 0.983008

[0.00s -> 20.00s] Deliver, request, fetch, green, red, blue, yellow, gray, black, small, medium, large, screwdriver, tool, object, wrench.

(misc) PS C:\Users\mrrya\Documents\MREN403\misc>

#note: 16 out of 16, perfect

Oct 19 – Run 3

#tiny model

Detected language 'en' with probability 0.984595

[0.00s -> 15.52s] The liver, request, fetch, green, red, blue, yellow, gray, black, small, medium, large,

[15.52s -> 19.68s] screwdriver, tool, object wrench.

#14 or 15 out of 16, not sure if the size tradeoff is worth it.

<many testing runs, tiny vs small model, beam_size, channel size 1 vs 3, sampling frequency, sound device>

Oct 20 – Troubleshooting example

#small model, using bee movie script as reference. Sounddevice python to record live microphone, but just recording a chunk and then processing it later in the script (pseudo live).

Starting Recording, start speaking in 1s

Now

Finished Recording

Detected language 'en' with probability 1.000000

[0.00s -> 5.00s] According to all known laws of aviation, there is no way a bee should be able to fly.

[5.00s -> 8.00s] Its wings are too small to get its fat little body off the ground.

[8.00s -> 12.00s] The bee, of course, flies anyway because bees don't care what humans think is impossible.

Oct 20 – Run #4

#imported datetime, announced faster-whisper model.

Execution time: 2026-10-20 00:53:45.084288

Using faster-whisper model: small

Starting Recording, start speaking in 1s

Now

Finished Recording

Detected language 'en' with probability 1.000000

[0.00s -> 5.00s] According to all known laws of aviation, there is no way a bee should be able to fly.

[5.00s -> 9.00s] Its wings are too small to get its fat little body off the ground.

[9.00s -> 14.00s] The bee, of course, flies anyway, because bees don't care what humans think is impossible.

Oct 26 – Run #5

#voice_stream.py

#Input stream from microphone, transcribe recorded segments in real time. Medium model taking too long though, as you can see. Transcription taking almost exactly the same time as recording, with beam size 5. Going to continue testing model sizes, beam sizes, and outputs.

(misc) PS C:\Users\mrrya\Documents\MREN403\misc> & c:/Users/mrrya/Documents/MREN403/misc/Scripts/python.exe c:/Users/mrrya/Documents/MREN403/misc/voice_stream.py

Execution time: 2026-10-26 21:49:31.277378

Using faster-whisper model: medium

Enter Stream

Audio duration: 5.018

Transcription took 5.03 seconds

[0.00s -> 4.60s] Hey, grab me the red-handled screwdriver off the board, will you?

Audio duration: 5.018

Transcription took 5.10 seconds

[0.00s -> 7.00s] Look on the pegboard and get me a tool, the bright blue screwdriver.

Audio duration: 5.018

Transcription took 4.79 seconds

[0.00s -> 7.00s] Can you fetch the green screwdriver off the rack for me?

Audio duration: 5.018

Transcription took 4.95 seconds

[0.00s -> 5.00s] Reach up to the rack and grab the yellow screwdriver with the magnetic screwdriver.

Audio duration: 5.018

Transcription took 5.00 seconds

[0.00s -> 2.00s] tip.

[2.00s -> 5.00s] Fetch me that blue screwdriver from the rack real quick.

Audio duration: 5.018

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Transcription took 4.95 seconds
[0.00s -> 7.00s]  Right there on the middle rack, grab this screwdriver with the red grip.
Audio duration: 5.018
Transcription took 5.06 seconds
[0.00s -> 4.98s]  Pass me the green screwdriver hanging right near the top of the
Audio duration: 5.018
Transcription took 4.91 seconds
[0.00s -> 2.74s]  board.
[2.74s -> 5.00s]  Deliver me the yellow screwdriver from the
Audio duration: 5.018
Transcription took 5.00 seconds
[0.00s -> 3.60s]  magnetic strip on the rack.
[3.60s -> 4.60s]  Bring me the green screwdriver.
Audio duration: 5.018
Transcription took 5.05 seconds
[0.00s -> 6.72s]  hanging on the main board. From the top row of the board
Audio duration: 5.018
Traceback (most recent call last):
  File "c:\Users\mrrya\Documents\MREN403\misc\voice_stream.py", line 50, in <module>
    segments = list(segments)
               ^^^^^^^^^^^^^^
  File "C:\Users\mrrya\Documents\MREN403\misc\Lib\site-
packages\faster_whisper\transcribe.py", line 1190, in generate_segments
    encoder_output = self.encode(segment)
                     ^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\mrrya\Documents\MREN403\misc\Lib\site-
packages\faster_whisper\transcribe.py", line 1400, in encode
    return self.model.encode(features, to_cpu=to_cpu)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

KeyboardInterrupt